# Topic 1.4: Population Density and Distribution



This revision pack covers **Topic 1.4: Population Density and Distribution** from the Cambridge IGCSE Geography 0460 specification (old syllabus, last exams 2026). It includes key definitions, a factors analysis, two contrasting case studies (Hong Kong and Chesterfield Inlet, Canada), exam practice questions and a full mark scheme.

| Syllabus learning objectives (0460) |
|---|
| • Define and calculate population density. |
| • Describe and explain the world's uneven population distribution. |
| • Identify and explain physical and human factors affecting population density. |
| • Use named case studies of a densely populated area and a sparsely populated area. |

## KEY DEFINITIONS

| | |
|---|---|
| **Population density** | The average number of people per square kilometre (km²) in an area. Formula: Population ÷ Area (km²). |
| **Population distribution** | How people are spread across an area — some places are densely populated, others are sparsely populated. |
| **Densely populated** | An area with a large number of people per km², e.g. Hong Kong (~6,700/km²). |
| **Sparsely populated** | An area with very few people per km², e.g. Chesterfield Inlet (~0.0005/km²). |
| **Dot map** | A type of map used to show distribution — each dot represents a set number of people. Clustered dots = dense; scattered dots = sparse. |

**Why does population density matter?** Governments use population density data to plan services such as schools, hospitals and transport. Areas of high density face pressure on resources; sparsely populated areas face challenges providing services across large distances.

## FACTORS AFFECTING POPULATION DENSITY

Population density is not random — it is shaped by both **physical** (natural) and **human** (socio-economic) factors. The same factor can attract people to one area while repelling them from another.

| Physical Factors | Human Factors |
|---|---|
| • Relief & altitude — low, flat land is easier to build on and farm; steep mountains are hostile. | • Economic activity — industrial zones, ports and commercial centres attract workers. |
| • Climate — mild, warm and wet climates support agriculture; extreme cold/heat deters settlement. | • Employment — areas with jobs attract migrants; rural areas with little work lose people. |
| • Water supply — rivers and lakes attract settlement; deserts and dry areas repel it. | • Transport — good road, rail and air links make an area accessible and attractive. |
| • Soil quality — fertile soils (e.g. river deltas) support farming and dense populations. | • Political stability — conflict and instability drive people away. |
| • Natural hazards — areas prone to earthquakes, volcanoes or flooding tend to be avoided. | • Historical factors — some cities have been centres of trade for centuries. |
| • Natural resources — mineral wealth, forests and fish stocks attract workers. | • Services — good schools, hospitals and infrastructure attract families. |

# Topic 1.4: Population Density and Distribution

**★ Exam tip: Physical vs Human**

- Always distinguish between physical and human factors — you will lose marks if you confuse them.
- Learn at least TWO examples for each factor type.
- For high-density areas, explain WHY factors attract people; for low-density areas, explain WHY they deter settlement.

## CASE STUDY A: HIGH DENSITY — HONG KONG, CHINA

### CASE STUDY: HONG KONG

*High-density urban area · ~6,700 people/km²*

*Fig. 1 — Victoria Harbour skyline viewed from Victoria Peak.*

*Fig. 2 — High-density residential blocks, Kowloon district.*

### Location and basic facts

- Located on the southern coast of China, bordering Guangdong Province.
- Total area: approximately 1,110 km² — about the size of Greater London.
- Population: approximately 7.4 million (2023 estimate).
- Population density: ~6,700/km² overall; Kowloon reaches over 40,000/km².
- Special Administrative Region (SAR) of China since 1997.

### Physical factors favouring high density

Despite its mountainous terrain — over 70% of Hong Kong is steep hills — the flat coastal lowlands and harbour provided an ideal base for settlement. The deep natural harbour attracted trade ships from the 19th century onwards. The subtropical climate, while bringing typhoons, also provides warmth and rainfall that support urban life.

- Deep natural harbour → ideal for trade and shipping.
- Flat coastal lowlands → available land for building (even though limited).
- Subtropical climate → warm year-round, supports diverse economic activity.
- Proximity to Pearl River Delta → fertile hinterland and fresh water access.

### Human factors favouring high density

- Global financial centre — one of Asia's leading stock exchanges; attracts multinational businesses.
- Exceptional transport links — international airport, MTR metro, cross-border rail to mainland China.
- Historical migration — waves of migrants from mainland China (1950s–1980s) swelled the population.
- Economic opportunity — low taxes, free-trade port status, strong banking sector create jobs.
- Limited land — scarcity forces high-rise development, increasing density on available flat land.

- Government investment — world-class healthcare, education and housing policy retain population.

## Consequences of high density

Hong Kong's extreme density creates both **advantages** and **challenges**:

| Advantages | Challenges |
|---|---|
| ✓ Efficient public transport (MTR serves >5 million journeys/day). | ✗ Housing unaffordable — median home costs over 20× average annual income. |
| ✓ Agglomeration economies — businesses benefit from clustering. | ✗ Overcrowding — some residents live in "cage homes" or subdivided flats <10 m². |
| ✓ Vibrant cultural life, restaurants, retail. | ✗ High land prices restrict manufacturing and drive inequality. |
| ✓ World-class hospitals and schools within short distances. | ✗ Traffic congestion and air pollution in urban core. |

## CASE STUDY B: LOW DENSITY — CHESTERFIELD INLET, CANADA

### CASE STUDY: CHESTERFIELD INLET

*Sparsely populated sub-Arctic community · ~0.0005 people/km²*

*Fig. 3 — Aerial view of Chesterfield Inlet settlement on the tundra.*

*Fig. 4 — Location map: Chesterfield Inlet, Nunavut, Canada.*

### Location and basic facts

- Located on the western shore of Hudson Bay, Nunavut Territory, northern Canada.
- Coordinates: approximately 63°N — deep within the sub-Arctic zone.
- Population: approximately 350–400 people (2021 Census).
- Area of surrounding region: vast tundra landscape spanning hundreds of km².
- Population density: among the lowest on Earth — fewer than 1 person per 2,000 km².
- Predominantly Indigenous Inuit community; official languages English, French and Inuktitut.
- Accessible only by air or, briefly in summer, by sea — no road connection.

### Physical factors causing low density

The extreme physical environment of Nunavut makes large-scale settlement almost impossible. These natural barriers have historically limited population growth and continue to do so today.

- Extreme cold — winter temperatures fall below –40°C; the ground is permafrost year-round.
- Short growing season — only 6–8 frost-free weeks per year; no commercial agriculture possible.
- Permafrost — permanently frozen ground makes construction extremely difficult and expensive.
- Remote location — situated on the western edge of Hudson Bay, far from major population centres.
- Tundra landscape — sparse vegetation (mosses, lichens) cannot support large animal herds for farming.
- Sea ice — Hudson Bay freezes for 8–9 months, cutting off sea access for most of the year.

### Human factors causing low density

- Lack of economic development — no significant mining, industry or manufacturing in the immediate area.
- Limited employment — most jobs are in government services, education, health care or traditional hunting.
- Poor infrastructure — no roads linking it to southern Canada; all goods arrive by expensive air freight.
- High cost of living — remote logistics mean food and materials cost 3–5× southern Canadian prices.
- Historical factors — Indigenous Inuit traditionally lived nomadically; permanent settlement is relatively recent.
- Out-migration — young people often leave for education and employment opportunities in larger cities.

### Consequences of low density

Sparse population creates both freedoms and significant hardships for residents:

| Advantages | Challenges |
|---|---|
| ✓ Vast wilderness and hunting grounds preserved for traditional Inuit practices. | ✗ High cost of food and goods — a basic grocery shop can cost 3–5× southern prices. |
| ✓ Low pollution, clean air, spectacular natural environment. | ✗ Limited access to health services — serious cases must be airlifted to Winnipeg or Ottawa. |
| ✓ Strong community bonds in small, tight-knit settlement. | ✗ Few educational choices beyond primary school; secondary students often board away. |
| ✓ Rich cultural heritage — Inuit traditions maintained. | ✗ Economic isolation — limited employment diversity leads to poverty and welfare dependency. |

**★ Comparing your two case studies**

- Hong Kong: physical harbour + human economic pull = extreme density.
- Chesterfield Inlet: extreme physical environment + limited human development = extreme sparsity.
- In the exam, always quote figures: "Hong Kong has ~6,700 people/km² compared to fewer than 1 person per 2,000 km² in the Chesterfield Inlet region."
- Link factors together — do not just list them. E.g. "Permafrost makes construction costly, which discourages investment, which limits jobs, which prevents population growth."

# Topic 1.4: Population Density and Distribution



## EXAM PRACTICE QUESTIONS

**Exam guidance**

- 1-mark questions: one correct fact or definition.
- 2-mark questions: two separate points, or one point with development.
- 3-mark questions: three points or one explained point with an example.
- 4-mark questions: two developed points, each with evidence or example.
- 7-mark questions: extended writing — use PEEL structure and refer to named examples.

### Question 1

**(a)** Define the term *population density*. **[1 mark]**

**(b)** State two physical factors that can cause an area to be sparsely populated. **[2 marks]**

**(c)** Explain how **climate** can affect population density. Refer to an example in your answer. **[3 marks]**

**(d)** Using a named example of a **densely populated area**, explain two human factors that have contributed to its high population density. **[4 marks]**

**(e)** "Physical factors are the most important reason why some areas are sparsely populated." Do you agree with this statement? Justify your answer with reference to a named example of a **sparsely populated area**. **[7 marks]**

**PEEL your answer** — Use this structure for 7-mark extended writing.

| P — Point | E — Evidence | E — Explain | L — Link back |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

## Question 2

**(a)** What is meant by the term *population distribution*? **[1 mark]**

**(b)** Give two human factors that attract people to live in a particular area. **[2 marks]**

**(c)** Explain how **economic activity** can lead to high population density. Use an example in your answer. **[3 marks]**

**(d)** Using a named example of a **sparsely populated area**, explain two physical factors that have contributed to its low population density. **[4 marks]**

**(e)** For a named **densely populated area** you have studied, explain how both physical and human factors have contributed to its high population density. Do you think physical or human factors are more important? Justify your answer. **[7 marks]**

**PEEL your answer** — Use this structure for 7-mark extended writing.

| P — Point | E — Evidence | E — Explain | L — Link back |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

## MARK SCHEMES

### Mark Scheme — Question 1

**(a)** [1 mark]

- The average number of people per square kilometre (km²) [1].
- Accept: Population ÷ Area.

**(b)** [2 marks]

- Award 1 mark for each correct physical factor, up to 2.
- Extreme cold / harsh climate [1]; mountainous/steep terrain [1]; permafrost/frozen ground [1]; desert/drought conditions [1]; flooding/swamp land [1]; infertile soils [1].

**(c)** [3 marks]

- Level 1 (1 mark): Simple statement. E.g. "Cold climates have few people."
- Level 2 (2 marks): Explained point. E.g. "Extreme cold makes farming impossible, so fewer people can survive there."
- Level 3 (3 marks): Explained point with named example. E.g. "In Chesterfield Inlet, Canada, temperatures fall below –40°C in winter, making agriculture impossible and construction on permafrost extremely costly, resulting in very low population density of under 1 person per km²."

**(d)** [4 marks]

- Award up to 4 marks: 2 marks per factor (1 mark for identifying factor + 1 mark for development/example).
- Named example required — e.g. Hong Kong, Singapore, Bangladesh.
- Factor 1: e.g. Economic opportunity — Hong Kong is a global financial centre attracting multinational businesses and millions of workers [1+1].
- Factor 2: e.g. Transport — Hong Kong's MTR, international airport and port make it highly accessible, encouraging further immigration and economic growth [1+1].
- Do not accept physical factors.

**(e)** [7 marks]

- Level 1 (1–2 marks): Simple, one-sided or list-style response. No example or weak example.
- Level 2 (3–4 marks): Some explanation of physical factors with a named example, but limited balance or underdeveloped argument.
- Level 3 (5–6 marks): Well-explained account of physical factors in a named example, with some consideration of human factors. Clear conclusion reached.
- Level 4 (7 marks): Balanced, well-structured argument using named example (e.g. Chesterfield Inlet). Physical factors (extreme cold, permafrost, remoteness) explained and supported with data. Human factors (lack of investment, poor infrastructure, out-migration) also considered. Justified conclusion — e.g. "Physical factors are the primary cause because the environment makes economic development almost impossible in the first place."
- Indicative content: Chesterfield Inlet — permafrost, –40°C winters, no roads, tundra, Hudson Bay sea ice; OR any valid sparse area with physical/human factor analysis.

### Mark Scheme — Question 2

**(a)** [1 mark]

- How the population is spread across an area [1].
- Accept: some places are crowded/dense, others are empty/sparse.

| **(b)** [2 marks] |
|---|
| • Award 1 mark for each correct human factor, up to 2. |
| • Employment/jobs [1]; good transport links [1]; political stability [1]; access to services (schools, hospitals) [1]; economic opportunity [1]; historical trade links [1]. |
| **(c)** [3 marks] |
| • Level 1 (1 mark): Simple statement. E.g. "Factories create jobs so people move there." |
| • Level 2 (2 marks): Explained point. E.g. "Economic activity creates employment, which attracts workers and their families, increasing population density." |
| • Level 3 (3 marks): Explained with named example. E.g. "Hong Kong's role as a global financial centre has attracted multinational corporations and millions of workers, contributing to an overall density of ~6,700 people/km².” |
| **(d)** [4 marks] |
| • Award up to 4 marks: 2 marks per factor (1 mark identifying + 1 mark development). |
| • Named example required — e.g. Chesterfield Inlet, Sahara Desert, Siberia. |
| • Factor 1: e.g. Extreme cold — temperatures below –40°C in winter make survival without expensive heating impossible [1+1]. |
| • Factor 2: e.g. Permafrost — permanently frozen ground makes building construction costly and limits land use to surface activities only [1+1]. |
| • Do not accept human factors. |
| **(e)** [7 marks] |
| • Level 1 (1–2 marks): Vague or list-style response. Weak or no named example. |
| • Level 2 (3–4 marks): Identifies physical and/or human factors with a named example. Limited depth; partial conclusion. |
| • Level 3 (5–6 marks): Good explanation of both factor types using Hong Kong. Supported with data/evidence. Conclusion present. |
| • Level 4 (7 marks): Thorough analysis of Hong Kong. Physical factors: deep harbour, flat coastal land, mild climate. Human factors: global financial centre, transport links (MTR, airport), historical migration, low taxes. Justified conclusion with a clear stance — e.g. "Human factors are more important because Hong Kong's physical land area is small and hilly, yet human economic decisions transformed it into one of the world's most densely populated places." |
| • Credit use of statistics: ~6,700 people/km² overall; Kowloon >40,000/km²; 7.4 million population. |